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Industrial Screw Picking via YOLO26 Parallel PSPE

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***Abstract*—Randomly piled screws pose a persistent challenge for robotic bin-picking. The combination of dense mutual occlusion, metallic specular reflections, and high aspect ratios tends to corrupt depth maps, undermining the reliability of conventional detection pipelines. To address this, we propose a Parallel Segmentation Pose Estimation (PSPE) architecture integrated within the YOLO26 framework. Existing approaches typically follow a serial design: a detected region is cropped and forwarded to a downstream pose estimator, so errors compound at each handoff. PSPE departs from this by running instance segmentation and keypoint regression as two independent branches directly on the full input image; their outputs are then reconciled through a confidence-weighted mask-voting scheme in post-processing. Screw geometry is parameterized by four keypoints—three distributed around the screw head and one at the tail centre. The head centre is computed as the coordinate-wise median of the three head keypoints, a choice that confers robustness when one keypoint drifts under partial occlusion. Orientation is then inferred from the head-to-tail vector. Experiments were conducted on a self-collected dataset comprising 2,229 annotated images across six screw categories. The segmentation branch (YOLO26n-seg) achieves mAP@50 $= 87.0\%$. PSPE increases the keypoint PCK@0.2 from 37.4% to 48.0% and reduces the head-centre localization error from 55.5 px to 47.1 px compared to the serial cascade baseline; at the same time, it lowers per-frame latency from 83.3 ms to 40.0 ms. Physical validation on a Dobot Magician v2 arm, with 150 trials in three stacking scenarios, shows a grasping success rate of 94.6%; thus, it is suitable for real-time small-part bin-picking using a low-cost RGB hardware system.**

***Index Terms*—Machine vision, part localization, deep learning, robotic grasping.**

I. Introduction

Bin Picking: The selection of a random component from an unstructured arrangement inside a box for automated industrial production equipment is currently unsolved. Screws are relatively difficult to use. Storage Area: Fasteners are in a pile, and many individual pieces overlap or are piled on top of one another; they are seen from all sides. Polished heads have a strong specular highlight and are thus very reflective. Regions and Corrupt Depth Readings. Recovery of a reliable grasp pose from such scenes has resisted direct methods [1]. The main industrial approach is to combine structured light or time-of-flight depth cameras with point cloud registration, and most commonly, the Iterative Closest Point (ICP) algorithm is used to obtain six-degree-of-freedom target poses. For small, shiny objects such as screws, this pipeline fails: the depth map returns noise and mirror-like artifacts, and the recovered pose is rarely accurate enough for gripper guidance [2]. High-fidelity 3D cameras have added to the price hike and are out of reach for small and medium-sized enterprises. Therefore, at present, deep-learning-based 2D vision has gained increasing attention as a relatively low-cost option. YOLO-series detectors have become the default choice for industrial visual inspection due to their high speed and relatively good accuracy, and few other architectures have been surpassed [3]. Recent advances in NMS-free end-to-end YOLO design have further reduced inference latency while maintaining competitive detection accuracy [4]. Moreover, small-object detection in cluttered industrial scenes remains an active research frontier, with multi-scale feature fusion strategies shown to substantially improve recall on sub-centimetre targets [5]. Based on the above, the present work has made the following contributions:

1) A PSPE parallel architecture that fully decouples instance segmentation from keypoint pose estimation, runs the two tasks in independent inference streams, and reconciles their outputs via a confidence-weighted mask-voting mechanism in post-processing to eliminate cascading error propagation by design.
2) A four-keypoint skeleton has been designed for screw geometry, and both the 3-D grasp centre and the in-plane orientation angle are obtained analytically from it.
3) A self-annotated dataset of 2,229 images covering six screw types (flange bolts, hex-head bolts, flange socket

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head cap screws, flat head screws, long pan head screws, and short pan head screws), accompanied by systematic ablation experiments.

4) End-to-end validation on a Dobot Magician v2 robotic arm, achieving a grasp success rate of 94.6%.

II. Related Work

A. Industrial Part Grasping

Research on automated bin picking stretches back to the 1980s, when template matching and handcrafted geometric descriptors dominated. The later availability of consumer depth cameras shifted the focus to point-cloud-based pipelines. Drost et al. [6] introduced the Point Pair Feature (PPF) descriptor for rigid-body detection, while the GraspNet framework [7] tackled the problem end-to-end. Both families, however, remain fragile on highly reflective metallic surfaces.

Pure RGB-based grasping has therefore attracted growing interest. GGCNN [8] and GR-ConvNet [9] predict pixel-level grasp-quality maps directly from convolutional feature maps, enabling efficient parallel-jaw grasp planning. Kumra et al. subsequently proposed GR-ConvNet v2 [10], achieving 98.8% accuracy on the Cornell dataset through improved multi-grasp detection at real-time speeds. For axisymmetric industrial parts specifically, Li et al. [11] demonstrated that combining instance segmentation with keypoint detection in a single convolutional network satisfies real-time requirements in stacked bin scenarios. These methods work well in single-category scenes but provide no instance-level semantic information and cannot localise geometrically functional points such as the screw-head centre—both of which are prerequisites for type-aware, precision-aligned assembly. The PSPE architecture proposed here retains the cost and robustness advantages of pure-RGB perception while adding instance-level semantics and functional keypoint localisation through a decoupled dual-stream network.

B. Instance Segmentation

Mask R-CNN [12] defined the canonical paradigm of joint detection and pixel-level mask prediction. SOLOv2 [13] later pushed the method towards real-time performance. From YOLOv8 onwards the YOLO family integrates instance segmentation via a lightweight mask decoder. Liu and Lv [14] further adapted the YOLO architecture specifically for bolt detection, demonstrating that task-specific lightweight modifications substantially improve both precision and frames-per-second on industrial fastener datasets. YOLO26 extends this by strengthening small-object detection and adopting an end-to-end dual-label-assignment strategy that replaces the traditional non-maximum suppression (NMS) post-processing step entirely, thereby removing a key source of latency overhead and sustaining throughput above 50 FPS on moderate GPU hardware.

C. Keypoint Pose Estimation

Keypoint-based pose estimation has reached maturity in the context of human-body and hand-pose prediction; representative works include HRNet [15] and YOLO-Pose [16]. YOLO-Pose integrates keypoint regression directly into the YOLO detector via an Object Keypoint Similarity (OKS) loss; a single forward pass then handles multiple instances at once. More recently, NMS-free YOLO variants have been extended to pose estimation: Meng et al. [17] proposed YOLOv10-pose and YOLOv9-pose for agricultural picking applications, while Ding et al. [18] demonstrated competitive PCK accuracy in human posture estimation at reduced latency relative to YOLOv8-pose baselines. Shifting this idea to industrial fasteners is not straightforward. Human-pose keypoints mark anatomical landmarks whose positions are constrained by body kinematics; screw keypoints, by contrast, must coincide with geometrically functional sites such as the head perimeter and shaft centre. Therefore, a new annotation scheme needs to be developed. Keypoint definitions, visibility labels and the order of linkage for these points all stem from the physical attributes of the target part and do not follow a universal convention.

D. Joint Segmentation and Pose Estimation

Methods that couple segmentation with pose estimation can be grouped by how tightly the two tasks share computation.

Serial cascade: A segmentation network crops a region of interest that is handed to a separate pose network. The structure is intuitive but susceptible to cascading error propagation—any mask imperfection is amplified by the downstream pose estimator [12].

Multi-task joint training: Mask R-CNN introduced shared-backbone, multi-branch inference to reduce error accumulation. Zhang et al. [19] extended this paradigm for 6D pose estimation via multi-task parameter sharing, narrowing accuracy gaps across objects of varying geometry on the LineMOD benchmark. For densely packed small industrial parts with heavy mutual occlusion, however, a single network's internal branches tend to produce cross-instance feature confusion and instance mismatches.

The PSPE architecture proposed here is designed to avoid both pitfalls. By running the two streams in strict independence and deferring all cross-task reasoning to a lightweight post-processing matching stage, it achieves the clean task separation of a serial pipeline without the associated error propagation, and the semantic richness of a joint model without its feature entanglement.

III. Method

A. Overall System Architecture

The PSPE pipeline runs through four stages in order: image acquisition, parallel inference (segmentation sub-network and pose-estimation sub-network), result matching, and grasp-parameter generation, as illustrated in Fig. 1.

Both sub-networks receive the full, unmodified input image and infer independently. The segmentation sub-network outputs per-instance binary masks together with class labels. The pose sub-network outputs, for each detected instance, four keypoint coordinates and their individual confidence scores. The matching module then associates these two output sets

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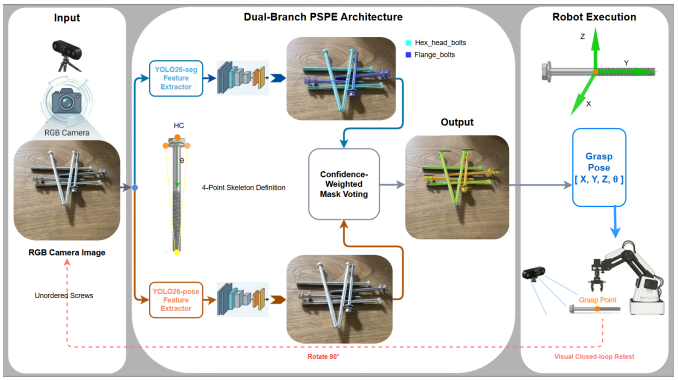


Fig. 1: Overall PSPE system architecture. Both sub-networks receive the full input image and infer independently; a confidence-weighted mask-voting algorithm reconciles their outputs in post-processing.

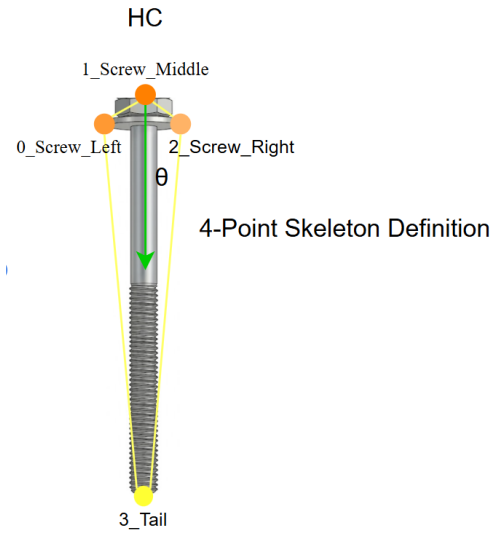


Fig. 2: Four-keypoint skeleton definition for screw geometry.

via the confidence-weighted mask-voting algorithm described in Section III-C.

B. Four-Keypoint Skeleton Definition

Four anatomically motivated keypoints are defined for each screw head and tail (see Fig. 2): **Screw_Left** (mid-point of the left edge of the screw head), **Screw_Middle** (top centre of the head), **Screw_Right** (mid-point of the right edge), and **Screw_Tail** (centre of the tail). During annotation, each keypoint is explicitly recorded as either visible or occluded.

Keypoints p_0, p_1, p_2 form a descriptive triangle over the screw head. The head centre HC is defined as their coordinate-wise median:

$$\mathrm{HC} = \mathrm{median}(\{p_0,\, p_1,\, p_2\})\,. \tag{1}$$

The median is preferred over the arithmetic mean because it suppresses the influence of any single poorly localised keypoint—if one head keypoint drifts due to partial occlusion, the median estimate is unaffected, whereas the mean would be pulled off-centre.

The in-plane screw orientation angle θ is derived from the direction vector connecting HC to the tail keypoint p_3:

$$\theta = \arctan 2(p_{3y} - \mathrm{HC}_y,\; p_{3x} - \mathrm{HC}_x)\,. \tag{2}$$

C. Confidence-Weighted Mask-Voting Matching

Let the pose sub-network produce N keypoint groups $\{KP_1, \ldots, KP_N\}$ and the segmentation sub-network produce M instance masks $\{Mask_1, \ldots, Mask_M\}$. For keypoint group KP_i and mask $Mask_j$, we define the confidence-weighted overlap score:

$$\mathrm{Score}(KP_i, Mask_j) = \sum_k w_k \cdot \mathrm{conf}_k \cdot \mathbf{1}[p_{i,k} \in Mask_j]\,, \tag{3}$$

where w_k is the per-keypoint weight ($w_k = 1.0$ for the three head keypoints; $w_k = 0.5$ for the tail keypoint, reflecting its higher occlusion probability), conf_k is the predicted confidence of keypoint k, and $\mathbf{1}[\cdot]$ is the indicator function that evaluates to 1 if point $p_{i,k}$ falls within the mask region and to 0 otherwise.

For each mask $Mask_j$ the keypoint group achieving the highest score is selected as its match; masks whose maximum score is zero are flagged as invalid instances. Although YOLO26's NMS-free single-stream design eliminates redundant detections within each task, it cannot resolve cross-model instance alignment between the segmentation and pose branches. The proposed voting algorithm effectively performs cross-task semantic deduplication in post-processing, filling that gap. Its time complexity is $O(N \times M)$; since both N and M are typically below 20 in practice, the computational overhead is negligible.

D. Grasp-Parameter Generation

For each valid screw, the system outputs: head-centre pixel coordinates (u_x, u_y); in-plane orientation angle $\theta \in (-180°, 180°]$; elevation angle $\alpha \in [0°, 90°]$ relative to the image plane; and screw class label.

When 3-D coordinates are required, the pixel location is back-projected using the camera intrinsic matrix and the depth value Z read from the RGB-D sensor:

$$X = \frac{(u_x - c_x)\, Z}{f_x}\,, \quad Y = \frac{(u_y - c_y)\, Z}{f_y}\,, \tag{4}$$

where (c_x, c_y) is the principal point and (f_x, f_y) are the focal lengths, both obtained by standard camera calibration. The orientation angle θ maps directly to the end-effector's rotation axis; the gripper is commanded to align perpendicularly to the screw axis so as to achieve the optimal constrained grasp. For vertical pick-and-place, only the 2-D orientation angle is needed at execution time; no 3-D pose calculation is carried out by the system online. Depth is determined by a single sensor-register read operation at the beginning of the arm's descent. Negligible cost.

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IV. Dataset Construction

A. Data Acquisition

Data collection was used to reproduce a typical scene in a factory for bin-picking tests. Fasteners went into the bin in whatever configuration they landed—no sorting, no deliberate arrangement. Images were taken with the camera fixed overhead, covering pile configurations from single-layer spreads up to four or five overlapping pieces. The denser stacks were a headache to annotate. Boundary calls on partially occluded heads kept coming back for re-review, and annotators disagreed more often than on the sparser piles. We kept them anyway. Removing high-clutter cases would have produced a cleaner dataset, but not a realistic one. To maximise diversity, the collection spanned multiple shooting distances (30–80 cm), four lighting conditions (natural daylight, artificial white light, side lighting, and mixed illumination), six background textures (corrugated cardboard surface, patterned marble table, wood-grain, metallic, industrial-mat, and plastic tray), and varying pile densities. Raw images were captured at 3048×3048 pixels with an Orbbec Gemini 2 RGB-D camera and labelled online via the Ultralytics annotation platform.

The dataset covers six screw categories—flange bolts, hex-head bolts, flange socket head cap screws, flat head screws, long pan head screws, and short pan head screws—comprising 2,229 annotated images in total (1,784 training, 445 validation). Compared with the initial two-category, 803-image collection, the expanded dataset introduces four additional screw types with shaft diameters ranging from 0.4 mm to 0.8 mm, broadening variation in object scale, head geometry, and surface reflectance. Data collection diversity was also increased: images now span a wider range of camera viewpoints, four lighting conditions (natural daylight, artificial white light, side lighting, and mixed illumination), six background textures (corrugated cardboard, patterned marble, wood-grain, metallic, industrial-mat, and plastic tray), and pile densities from isolated single screws to heavily stacked configurations. Approximately 40 validation images contain no visible screws and are retained specifically to assess false-positive robustness. Representative samples and annotation overlays are shown in Fig. 3.

B. Annotation Protocol

Annotation was performed at two levels on the same image batch: instance-segmentation masks (polygon outlines of each visible screw contour) and keypoint poses (following the skeleton defined in Section III). Keypoint visibility was recorded as *visible* (confidence label 2) or *occluded* (confidence label 1).

To ensure label consistency the following rules were enforced: (i) head keypoints must lie on the screw-head contour boundary; (ii) screws completely hidden by other objects are skipped entirely; (iii) keypoint order (Left→Middle→Right→Tail) is invariant to screw orientation, with left/right determined from the camera's viewpoint.

Dataset statistics are given in Table I.



(a) Dataset sample images—various screw configurations.

(b) Additional samples with annotation overlays.

Fig. 3: Dataset samples covering all six screw categories under diverse lighting conditions, background textures, and pile densities.

TABLE I: Dataset Statistics

Category	Total	Train	Val	Annotations
Flange bolts	500	400	100	1536
Hex head bolts	589	471	118	1382
Flange Socket Head Cap Screw	356	285	71	359
Flat Head Screw	92	74	18	93
Long Pan Head Screw	345	276	69	345
Short Pan Head Screw	347	278	69	347
Total	**2229**	**1784**	**445**	**4062**

V. Experiments

A. Experimental Setup

All experiments were run on a workstation equipped with an NVIDIA GeForce RTX 4050 GPU (6 GB VRAM) under Windows 11 Pro, with PyTorch 2.1 and Ultralytics 8.x. The two sub-networks were trained under differentiated configurations reflecting their respective task demands. For the segmentation sub-network, YOLO26n-seg (nano) was fine-tuned at 1280×1280 input resolution for 300 epochs—a high-resolution setting chosen specifically to capture fine-grained screw-edge details. The pose sub-network underwent a two-stage training strategy: a YOLO26s-pose (small) backbone was first trained as the base model; the resulting weights then initialised a second training stage at 640×640 resolution for 100 epochs, noticeably improving robustness across the diverse occlusion conditions present in our dataset. Both stages employed the AdamW optimiser with an initial learning rate of 0.01 and linear learning-rate decay.

B. Evaluation Metrics

Segmentation performance is reported via standard instance-segmentation metrics mAP^{M}_{50} and $\text{mAP}^{M}_{50:95}$. Pose estimation is evaluated with OKS-based mAP^{P}_{50} and PCK@0.2 (the fraction of keypoints falling within $0.2\times$ the head-region scale of their ground-truth locations). System-level performance in physical trials is captured by the Grasp Success Rate (GSR). We additionally report Head-Centre Error (HCE, in pixels) and Instance-level Classification and Matching Accuracy (ICMA).

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TABLE II: Instance Segmentation Results on the Validation Set

Model	Category	mAP_{50}^{M}	$\text{mAP}_{50:95}^{M}$	Prec.	Rec.
YOLO26n-seg	Flange	94.0%	59.4%	95.2%	85.6%
YOLO26n-seg	Hex-head	91.6%	67.2%	94.9%	84.7%
YOLO26n-seg	Flange Socket	95.5%	82.1%	97.6%	95.8%
YOLO26n-seg	Flat Head	59.3%	47.0%	54.6%	75.0%
YOLO26n-seg	Long Pan Head	87.3%	53.6%	87.7%	83.3%
YOLO26n-seg	Short Pan Head	94.3%	74.8%	90.3%	89.7%
YOLO26n-seg	**Overall**	**87.0%**	**64.0%**	**86.7%**	**85.7%**

C. Segmentation Performance

Table II reports validation-set segmentation results. YOLO26n-seg achieves an overall mAP_{50}^{M} of 87.0%, demonstrating reliable discrimination across all six screw categories. At 40.0 ms per frame (25 FPS), the system's throughput stems primarily from YOLO26's NMS-free end-to-end design, which bypasses the CPU-side suppression step that older YOLO variants require.

D. Pose Estimation and Architecture Comparison

Table III compares three architectural configurations: (A) serial cascade—pose inference on segmentation-cropped ROIs; (B) standalone full-image pose inference; (C) the proposed PSPE parallel architecture.

Config. C records a lower mAP_{50}^{P} than Config. B (55.6% vs. 67.0%). This discrepancy does not reflect model degradation. Because YOLO26 is NMS-free, the reduction cannot be blamed on over-suppression; it instead traces directly to the mask-gating mechanism that is central to PSPE. In densely occluded scenes, the pose branch can predict keypoints for partially buried screws that the segmentation branch cannot cover with a complete mask. PSPE intentionally flags such keypoint groups as high-risk candidates and excludes them from the grasp list; this is a deliberate conservative measure because the excluded screws are precisely those for which accessibility has not been verified by a segmentation mask.

The figures are as follows. HCE drops to 47.1 px; ICMA reaches 97.9% without any auxiliary classification branch; and at 40.0 ms per frame, the system runs at approximately 25 FPS, which is about half the latency of the serial baseline (83.3 ms) and well within the requirements for real-time deployment.

Why Moderate PCK is Sufficient. PCK@0.2 is relative to the head-region scale of each individual screw and not to image width. At our working distance of 292.5 mm, that threshold is about 1–2 px per screw head, and this is already tighter than what the gripper needs. Dobot v2 jaws open to 28 mm, and the shaft diameters in our dataset are 0.4–0.8 mm. By working with geometry, the allowable lateral offset before a grasp failure is about $\pm$13.7 mm per side, which corresponds to approximately 96 pixels at the working distance. Our measured HCE of 47.1 px (6.7 mm) is well within that margin. PCK@0.2 is a pixel-level bar that is much finer than the mechanical bar actually imposed by the jaw. For smaller fasteners or sub-millimetre tolerances, the gap would be reduced, and at this point, higher keypoint accuracy would begin to be necessary; it is a real constraint that cannot be ignored.

TABLE III: Keypoint Detection Performance Under Different Architectural Configurations

Cfg.	Description	mAP_{50}^{P}	PCK@0.2	HCE (px)	ICMA	Lat. (ms)
A	Serial: Seg→crop→Pose	31.0%	37.4%	55.5	98.6%	83.3
B	Standalone full-image Pose	67.0%	46.8%	53.7	N/A	24.9
C	PSPE (ours)	55.6%	48.0%	47.1	97.9%	40.0

TABLE IV: Ablation Results. "Dedup."=mask-gating deduplication; "C.wt."=confidence weighting; "T.dw."=tail down-weighting.

Exp.	Dedup.	C.wt.	T.dw.	mAP_{50}^{P}	Dup. rate	ICMA	HCE	PCK@0.2
E1	✗	✗	✗	67.0%	24.0%	97.3%	53.7	46.8%
E2	✓	✗	✗	54.5%	5.6%	97.9%	51.1	47.9%
E3	✓	✓	✗	55.4%	5.7%	98.0%	47.3	47.7%
E4	✓	✓	✓	55.6%	5.7%	97.9%	47.1	48.0%

E. Ablation Study

Table IV shows the contents of each algorithm.

1) Mask Gating: Deduplication and Purification: The E1–E2 contrast is used to read off mask-gated deduplication locally. No spatial masking (E1) is used, and thus activations are duplicated in areas of high overlap. Climb to 24.0%. The actual loop usually has this redundancy of multiple grasp commands and occasional arm interference. E2 adds a greedy one-to-one mask gate before finalisation of boxes; the duplicate rate is reduced to 5.6%, and after removing keypoints without mask support, HCE decreases from 53.7 px to 51.1 px. For the same reason, mAP falls from 67.0% to 54.5%. Occluded instances that do not receive a mask are excluded by design; a decrease is not to be interpreted as weaker localisation.

2) Soft-Weighting Strategies: E2 provides the hard spatial boundary; E3 and E4 are then selected within it. Confidence weighting (E3) aims to adjust the prediction corresponding to the acceptance threshold, and by recovering these borderline detections, mAP increases from 54.5% to 55.4%. Tail down-weighting (E4) is based on the empirical observation that screw tails are often truncated at the edge of a mask or obstructed by adjacent fasteners. The whole rise of the indicator is relatively small; in the case of severe tail occlusion or substantial overlap among many screws, the highest value reached by HCE is around 47.1 pixels.

3) Summary: Duplicate Detections are less than 6%; HCE lands at 47.1 px. Neither of the above results needs an additional classification branch. The network remains small, and latency is still relatively low because no additional inference paths have been added.

F. End-to-End Grasping Experiments

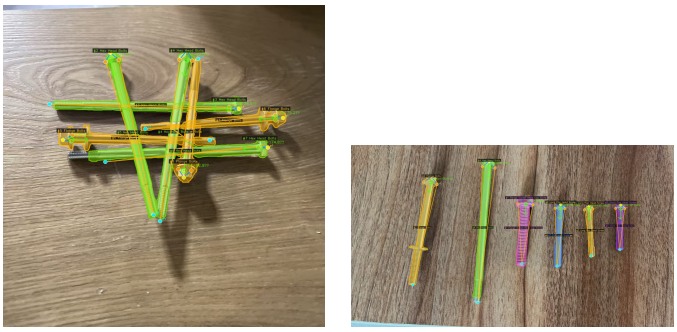
A Dobot Magician v2 robotic arm was used for physical verification. Three scenarios were tested, and in each case, 50 grasping attempts were made: (1) single screw, no occlusion; (2) multiple screws, no stacking; (3) multiple screws with random stacking and partial occlusion. The results are in Table V.

Representative output visualisations are shown in Fig. 4. The total GSR of the system is 94.6%. A 90° rotation retry mechanism is used for a small number of cases in which the

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TABLE V: End-to-End Grasping Experiment Results

Scenario	Att.	1st-try	2nd-try	Fail	GSR
Single, no occl.	50	47	2	1	98.0%
Multi., no stacking	50	45	3	2	96.0%
Multi., w/ stacking	50	41	4	5	90.0%
Total	**150**	**133 (88.6%)**	**9 (6.0%)**	**8 (5.3%)**	**94.6%**



(a) Multiple overlapping screws.

(b) Segmentation and keypoint overlay.

Fig. 4: Representative output visualisations from the PSPE algorithm showing instance masks and detected keypoints on stacked screws.

gripper is parallel to the screw axis on the first attempt, and most of these marginal failures are thus resolved. Failures occurred in the most extreme stacking cases; all head keypoints were completely occluded, and orientation estimation was therefore unreliable.

G. Comparison with Existing Methods

To provide a more practical comparison, GGCNN [8] and GR-ConvNet [9] were also reproduced and retrained on our screw data under the same conditions. As no paired depth maps were saved during data collection, RGB-only input was used, and pseudo-depth was derived from the instance segmentation masks for format compatibility. GGCNN achieves a grasp success rate of 61.9% at approximately 447 FPS, and GR-ConvNet reaches 68.3% at approximately 329 FPS. The updated GR-ConvNet v2 [10] raised accuracy to 98.8% on standard benchmarks, yet its pixel-wise grasp map still provides no instance-level semantics or functional keypoint outputs, limiting applicability to category-aware assembly. While both reproduced baselines benefit from high throughput, neither produces instance-level semantic labels nor functional keypoint localisation, limiting their applicability to category-aware, precision-aligned assembly tasks. PSPE outperforms both baselines by a substantial margin (94.6% GSR) while additionally providing screw category, head-centre coordinates (HCE = 47.1 px), and in-plane orientation angle—outputs that are prerequisites for type-aware bin-picking. The higher inference rate of GGCNN is attributable to its lightweight all-convolutional design; PSPE's throughput of approximately 25 FPS nonetheless satisfies the real-time requirements of the Dobot Magician v2 platform. GraspNet [7] and PPF [6] are listed for context. Each was tested on a different dataset and platform, so their FPS numbers are rough order-of-magnitude indicators only. As summarised in Table VI, PSPE is the only RGB-only method in this comparison that simultaneously achieves good small-target handling and competitive keypoint accuracy.

TABLE VI: Comparison with Representative Existing Methods

Method	Modality	Depth	Small-tgt	FPS	Kpt. acc.
GraspNet [7]	RGB-D	Yes	Moderate	∼10	N/A
PPF [6]	Depth	Yes	Poor	∼5	N/A
GGCNN+(reproduced) [8]	RGB+	No	Moderate	∼447	N/A
GR-ConvNet+(reprod.) [9]	RGB+	No	Good	∼329	N/A
Serial Seg+Pose (baseline)	RGB	No	Poor	∼12	37.4%
PSPE (ours)	RGB	No	Good	**∼25**	**48.0%**

VI. Conclusion

Running instance segmentation and keypoint regression as two independent full-image streams—joined only at post-processing via a mask-voting step—turns out to sidestep the two failure modes that serial and joint-training architectures each suffer from. Five results support this.

1) **Speed and accuracy both improve over the serial cascade.** PCK@0.2 goes from 37.4% to 48.0%, HCE from 55.5 px to 47.1 px, and per-frame latency from 83.3 ms to 40.0 ms—a 2× speedup at higher geometric precision.
2) **Mask-gating cuts duplicate detections without an extra branch.** The greedy one-to-one matching step brings the duplicate rate from 24.0% down to 5.7% and holds ICMA at 97.9%, adding no additional network parameters.
3) **Broader data coverage generalises across screw types.** Moving from 803 images over two categories to 2,229 images over six (shaft diameters 0.4–0.8 mm) confirms that PSPE is not tuned to a single head geometry or surface finish.
4) **Four keypoints are enough to pin down grasp pose.** The sparse head-triangle plus tail-centre layout provides a 2-D grasp centre and orientation sufficient for the vertical pick-and-place task without requiring full 3-D pose recovery; tail down-weighting helps address the problem of a shaft being hidden by nearby screws.
5) **Practical application tests on the low-cost arm.** In 150 pick attempts on a Dobot Magician v2, the total grasping success rate is 94.6%, and this includes the most difficult stacking case.

Several extensions remain open. The RGB-D depth channel can be fused in the parallel branches at inference time. This change will no longer require a separate hand-eye calibration stage before deployment. Training may also combine mask-boundary consistency and keypoint gradients in a single joint loss. Contour error and joint drift would then be penalized together rather than individually. Besides screws, the same pipeline could be applied to bearing rings, irregular clips, and other small parts with silhouettes that deviate more from the screw templates used in this study. For fasteners below M2, higher-resolution keypoint labels will be increasingly required. At this scale, the difference between reported HCE and the permitted placement tolerance is already relatively small.

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REFERENCES

[1] K. Kleeberger, R. Bormann, W. Kraus, and M. F. Huber, "A survey on learning-based robotic grasping," *Current Robotics Reports*, vol. 1, no. 4, pp. 239–249, 2020.

[2] E. Brachmann *et al.*, "Learning 6D object pose estimation using 3D object coordinates," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 536–551.

[3] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLO," 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

[4] A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, J. Han, and G. Ding, "YOLOv10: Real-time end-to-end object detection," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 37, 2024.

[5] Q. Feng, X. Xu, and Z. Wang, "Deep learning-based small object detection: A survey," *Math. Biosci. Eng.*, vol. 20, no. 4, pp. 6551–6590, 2023.

[6] B. Drost, M. Ulrich, N. Navab, and S. Ilic, "Model globally, match locally: Efficient and robust 3D object recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2010, pp. 998–1005.

[7] H.-S. Fang, C. Wang, M. Gou, and C. Lu, "GraspNet-1Billion: A large-scale benchmark for general object grasping," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 11444–11453.

[8] S. Kumra, S. Joshi, and F. Sahin, "Antipodal robotic grasping using generative residual convolutional neural network," in *IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2020, pp. 9626–9633.

[9] S. Kumra and C. Kanan, "Robotic grasp detection using deep convolutional neural networks," in *IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2017, pp. 769–776.

[10] S. Kumra, S. Joshi, and F. Sahin, "GR-ConvNet v2: A real-time multi-grasp detection network for robotic grasping," *Sensors*, vol. 22, no. 16, p. 6208, 2022.

[11] Y. Li, F. Guo, M. Zhang, S. Suo, Q. An, J. Li, and Y. Wang, "A novel deep learning-based pose estimation method for robotic grasping of axisymmetric bodies in industrial stacked scenarios," *Machines*, vol. 10, no. 12, p. 1141, 2022.

[12] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 2961–2969.

[13] X. Wang *et al.*, "SOLOv2: Dynamic and fast instance segmentation," *Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 17721–17732, 2020.

[14] Z. Liu and H. Lv, "YOLO Bolt: A lightweight network model for bolt detection," *Sci. Rep.*, vol. 14, p. 656, 2024.

[15] J. Wang *et al.*, "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, 2020.

[16] D. Maji, S. Nagori, M. Mathew, and D. Poddar, "YOLO-Pose: Enhancing YOLO for multi person pose estimation using object keypoint similarity loss," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, 2022, pp. 2637–2646.

[17] Z. Meng, X. Du, R. Sapkota, Z. Ma, and H. Cheng, "YOLOv10-pose and YOLOv9-pose: Real-time strawberry stalk pose detection models," *Comput. Ind.*, vol. 165, p. 104231, 2024.

[18] J. Ding *et al.*, "Research on human posture estimation algorithm based on YOLO-Pose," *Sensors*, vol. 24, no. 10, p. 3036, 2024.

[19] L. Zhang, X. Zhou, J. Liu, C. Wang, and X. Wu, "Instance-level 6D pose estimation based on multi-task parameter sharing for robotic grasping," *Sci. Rep.*, vol. 14, p. 7801, 2024.

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